

Classification of Certain Vertebral Degenerations Using MRI and Deep Learning

Authors:	Jiyo S Athertya, G Saravana Kumar
Publication / Venue:	Research article
Year:	2021
DOI:	10.1088/2057-1976/ac00d2
Primary Source URL:	https://pubmed.ncbi.nlm.nih.gov/33984847/
Inventory Category:	Adjacent lumbar MRI AI
Comparability / Priority:	Background Priority: Background

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Deep-learning classification Lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Modic changes / vertebral degeneration
Model / AI Method:	Deep-learning system
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Shows the wider degenerative lumbar MRI AI landscape beyond stenosis and discs.

Abstract / Summary

Abstract Background and Objective: This article describes a fully automatic system for classifying various spinal degenerative phenotypes namely Modic changes, endplate defects and focal changes which are associated with lower back pain. These are obtained from T1/T2 Magnetic Resonance Imaging (MRI) scans. Lower back pain is a predominantly occurring ailment, which is prone to have various roots including the anatomical and pathophysiological aspects. Clinicians and radiologist use MRI to assess and evaluate the extent of damage, cause, and to decide on the future course of treatment. In large healthcare systems, to circumvent the manual reading of various image slices, we describe a system to automate the classification of various vertebral degeneracies that cause lower back pain. **Methods:** We implement a combination of feature extraction, image analysis based on geometry and classification using machine learning techniques for identifying vertebral degeneracies. Image features like local binary pattern, Hu's moments and gray level co-occurrence matrix (GLCM) based features are extracted to identify Modic changes, endplate defects, and presence of any focal changes. A combination of feature set is used for describing the extent of Modic change on the end plate. Feature sensitivity studies towards efficient classification is presented. A STIR based acute/chronic classification is also attempted in the current work. **Results:** The implemented method is tested and validated over a dataset containing 100 patients. The proposed framework for detecting the extent of Modic change achieves an accuracy of 85.91%. From the feature sensitivity analysis, it is revealed that entropy based measure obtained from gray level co-occurrence matrix alone is sufficient for detection of focal changes. The classification performance for detecting endplate defect is highly sensitive to the first 2 Hu's moments. **Conclusion:** A novel approach to identify the allied vertebral degenerations and extent of Modic changes in vertebrae by exploiting image features and classification through machine learning is proposed. This shall assist radiologists in detecting abnormalities and in treatment planning.